

10 Production-Grade LangChain Projects (Finance & Commercial Vehicle Focus)

Below is the structured summary of 10 production-grade LangChain projects covering 90% of core concepts.

Project 1: Smart Invoice Parser for Commercial Vehicle Fleets (Easy)

Covers: Document Loading, Text Splitting, Embeddings, Vector Stores, Basic RAG

Problem: Manual invoice entry causes delays and errors.

Impact: Processing time reduced 6 hours → 10 mins.

Tech Stack: LangChain (PyPDFLoader, FAISS), Streamlit, Docker.

Solution: Parse PDF → embed → retrieve → structured JSON output.

Project 2: Driver Behavior Chatbot using Telematics Logs (Easy–Medium)

Covers: Conversational RAG, Memory.

Problem: Managers can't analyze driver behaviour quickly.

Impact: 30% reduction in unsafe incidents.

Tech Stack: CSVLoader, Chroma, FastAPI.

Solution: Embed telematics logs → conversational insights.

Project 3: Predictive Maintenance Agent (Medium)

Covers: Agents, Tools, ML Integration.

Problem: Unplanned breakdowns cost █50K–2L.

Impact: 40% downtime reduction.

Tech Stack: Agents, Pinecone, Scikit-learn.

Solution: Sensor analysis + maintenance recommendations.

Project 4: Finance Compliance Checker (Medium)

Covers: Metadata-based Retrieval, Hybrid Search.

Problem: Missing RC/PUC/Insurance leads to loan rejections.

Impact: 30% → 8% rejection drop.

Tech Stack: SelfQueryRetriever, Weaviate.

Solution: Hybrid RAG + JSON compliance summary.

Project 5: Multi-Document RAG for Finance Reports (Medium–Advanced)

Covers: MultiQuery, Compression, Reranking.

Problem: Analysts take 4+ hrs analysing reports.

Impact: 80% faster insights.

Tech Stack: MultiQueryRetriever, CohereRerank.

Solution: Ensemble retrieval + compressed summarisation.

Project 6: Autonomous Finance Agent for Cost Optimization (Advanced)

Covers: ReAct Agents, Tool Calling, Async.

Problem: Need optimal fuel/toll/route allocation.

Impact: 12–18% cost reduction.

Tech: Maps API, Toll API, OR-Tools, Redis.

Solution: Multi-step reasoning with tools.

Project 7: Production RAG Pipeline with Monitoring (Advanced)

Covers: LangSmith, Evaluation, CI/CD.

Problem: Answer drift & SLA issues.

Impact: 99.5% accuracy SLA.
Tech Stack: Kubernetes, Prometheus, GitHub Actions.
Solution: Monitoring + auto-evaluation + retraining.

Project 8: Multimodal CV Finance Assistant (Advanced)

Covers: GPT-4V, OCR, Tool Use.
Problem: Fraud in odometer/chassis verification.
Impact: 95% fraud reduction.
Tech: MultiModalChain, Pytesseract.
Solution: Compare images + documents.

Project 9: Autonomous Fleet Finance Copilot (Advanced–Production)

Covers: Full Stack, Agents, RAG, Tools.
Problem: CFO needs unified financial insights.
Impact: █2Cr+ profit recovery.
Tech: FastAPI, Pinecone, Celery, Kubernetes.
Solution: End-to-end AI finance copilot.

Project 10: Decentralized Finance Insights for CV Lenders (Ultra Advanced)

Covers: Web3, Async Agents.
Problem: Unbanked drivers lack credit scoring.
Impact: Loan eligibility increased 40%.
Tech: web3.py, Blockchain APIs.
Solution: On-chain data → credit score.
